Pablo Fernando Badilla Torrealba

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Professional Summary

Tech Lead and Machine Learning Engineer with more than 7 years articulating, designing, and bringing to production AI, data, and software solutions for banking, retail, academia, and the public sector. I lead technical teams, define scalable architectures, and turn generative AI prototypes, predictive models, data pipelines, and research capabilities into high-impact products, coordinating with clients and upholding software development best practices and end-to-end execution.

Work Experience

Data and Artificial Intelligence Initiative (IDIA) – Universidad de Chile

Santiago, Chile

Tech Lead

Apr 2026 – Present

- **Project coordination and technical evaluation**: Coordination and technical evaluation of multiple AI and data projects for the faculty, research centers, and private companies. Outreach activities: talks on MLOps and AI-assisted programming.
- **Technical evaluation for FONDEQUIP application**: Evaluation of GPU requirements for LLM inference and neural network training, and collection of quotations.
- **AI-based cultural indicators management**: Implementation and technical leadership of a system for the automated extraction of ~50 cultural indicators for the Undersecretariat of Culture. Development of an advanced RAG-based pipeline with automated validation, together with a full-stack management app and public dashboard.
- **Agentic platform for kinesiology routines**: Design and development of an agentic platform that generates routines based on kinesiology recommendations.
- **Advisory for a shipping company**: Advisory on implementing an agentic system for commercial queries about the Chilean export market, focused on commercial decision-making.
- **AWS Breaking Barriers Hackathon**: Coordination of the Tornados project, a challenge from the Center for Climate and Resilience Research (CR2), and AI mentoring at the event in Chile.

Banco Falabella

Santiago, Chile

Data Scientist

Aug 2024 – Apr 2026

- **Products and benefits**: Chat with AI agents for queries about products, benefits, operations, and branches in Chile, Colombia, Mexico, and Peru. +100k monthly interactions and an 80% reduction in costs, as well as improvements in latency and user experience. Management of relationships with internal clients in all countries in collaboration with the Digital Channels team.
- **Transaction dispute**: Development and launch of a transaction dispute process for fraud in unreconciled transactions. First AI initiative published by the bank, opening the doors to more complex projects and an AI-based strategy. Collaboration with the Digital Channels team.
- **Concurrent conversation server library**: Design and implementation of a library based on concurrent and distributed algorithms to coordinate conversations across different AI use cases. Foundation for new projects based on conversational AI.
- **Mentoring**: Mentoring the team in conversational agent design, backend development, and AI-assisted software development.

Walmart Chile

Santiago, Chile

Machine Learning Engineer

Apr 2023 – Apr 2024

- **Call Center demand forecasting**: Development and deployment of a Machine Learning-based time series model for daily Call Center demand forecasting. Reduction of error from 60% to 20%. Evaluation through A/B testing and shadow deployment with ML metrics and cost-related KPIs. Milestone planning and joint presentation of progress and results to internal clients from the operations team together with the PO.
- **Queue wait-time simulation**: Optimization of a project to measure wait times according to format (Líder, Express, aCuenta, Central Mayorista) and checkout type (MM1 and MMS) using queuing theory. Substantial refactoring of the system's architecture and performance.
- **Customer Lifetime Value (CLV)**: Reimplementation of the Customer Lifetime Value model for the Marketing area, enabling personalized campaigns according to the detected customer segment.

DashAI Santiago, Chile

Tech Lead Jan 2023 – Jun 2024

- **Technical leadership and coordination**: Technical leadership of a team of five developers (thesis students from Universidad de Chile), team management, facilitation of ceremonies, and definition of technical and integration requirements. Repository: GitHub repository
- **Architecture, development, and technology transfer**: Construction of core components – plugin registry, dependency injection, subproject integration, testing, initial frontend, and CI –, together with the FONDEF 2023 application, the generation of a business model, and the design of a technology transfer strategy.

BCI Santiago, Chile

Data Scientist Feb 2023 – May 2023

- **Features for credit risk assessment**: Creation of features based on customers' transactional behavior to explore improvements in credit risk assessment models.

National Center for Artificial Intelligence – CENIA Santiago, Chile

Machine Learning Engineer Jan 2022 – Jan 2023

- **End-to-end ML model for VRP**: Design and deployment of a Machine Learning model that allows user preferences to be incorporated into the calculation of solutions to the Vehicle Routing Problem (VRP). Communication with clients, presentation of progress, milestone and delivery management in collaboration with the Optimization team. Delivery of a complete package ready for deployment in internal systems.

Universidad de Chile Santiago, Chile

Part-time Lecturer Mar 2021 – Jul 2024; Jul 2026 – Sep 2026

- **Teaching in Data Science**: Teaching in the Scientific Programming Laboratory for Data Science of the Master in Data Science (MDS) and in the Data Handling and Extraction in Python course of the Diploma in Python Applied to Data Science, both at Universidad de Chile.
- **AI agents course**: Taught a course on AI agents for the DCC's PEC, jointly with IDIA, from basic concepts to advanced topics: RAG, multi-agent systems, evaluation, and more. 100+ students across 36 classes.

Kwiyx SPA Santiago, Chile

Full-stack Developer Apr 2020 – Jan 2021

- **Full-stack application development**: Creation of the front-end of a real estate investment application and development of a human resources management system for hospitals during the pandemic, based on shift optimization.

Freelance Developer Santiago, Chile

Data Scientist and Full-stack Developer Jul 2019 – Jan 2021

- **Data and NLP projects**: Construction of a topic detection system using Transformer embeddings to analyze the 2019 participatory meeting of "Lxs 400" around the new Constitution, and development of an OCR + Regex-based system to extract text from automotive credit promissory notes.

Acción Circular Santiago, Chile

Full-stack Developer Apr 2019 – Jul 2019

- **Recyclable waste management**: Design and implementation of software to manage waste collection.

Projects

- **Word Embeddings Fairness Evaluation Framework (Jan 2020 – Jun 2025)**: Leadership and development of a library for measuring and mitigating gender, ethnic, and religious biases in word embedding models (precursors to language models). 2 published papers, 182 GitHub stars. Repository: GitHub | Documentation: Documentation

Education

Computer Engineering (Civil), Universidad de Chile, Santiago, Chile 2013 – 2019

Master of Computer Science, Universidad de Chile, Santiago, Chile 2019 – 2020

Publications

- **WEFE: A Python Library for Measuring and Mitigating Bias in Word Embeddings**: Pablo Badilla, Felipe Bravo-Marquez, María José Zambrano, Jorge Pérez. *Journal of Machine Learning Research*, Vol. 26, No. 156, pp. 1–6, 2025. JMLR article
- **WEFE: The Word Embeddings Fairness Evaluation Framework**: Pablo Badilla, Felipe Bravo-Marquez, Jorge Pérez. *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI-20)*, pp. 430–436, 2020. DOI

Skills and Tools

- **Leadership and architecture**: Technical leadership, software architecture, scalable systems design, mentoring, technical project management, and technology transfer
- **Generative AI and LLMs**: LLMs, RAG, conversational agents, multi-agent systems, MCP (Model Context Protocol), LLM evaluation, vector databases, LangChain, LangGraph, Langfuse (LLM observability), and OpenAI and Gemini APIs
- **Data Science and Machine Learning**: Python, NumPy, Pandas, SQL, PostgreSQL, BigQuery, Spark, Databricks, Scikit-Learn, XGBoost, LightGBM, PyTorch, and Hugging Face
- **Backend and software development**: FastAPI, SQLAlchemy, Pydantic, Asyncio, JavaScript, systems integration, testing, and full-stack development
- **Cloud, MLOps, and infrastructure**: Google Cloud Platform, AWS, Vertex AI, Docker, Kubeflow, Git, GitHub Actions, GitLab CI, and MLOps practices
- **Tools and methodologies**: Scrum, CI/CD, and software development best practices

Additional Information

- **Graduated with Highest Distinction in Engineering and Master's degree.**
- **Languages**: Native Spanish, B2 English TOEFL-ITP
- **Outstanding Student in 2016 and 2018**: Faculty of Physical and Mathematical Sciences, Universidad de Chile.
- **Undergraduate Thesis Advisor**: Implementation of a bias mitigation module in word embedding models in WEFE.
- **Guest Lecturer**: Talks on Machine Learning Operations and for various undergraduate and graduate courses at Universidad de Chile.
- **Microsoft College Code Competition**: Third place in the Microsoft College Code Competition 2018.
- **Paper Reviewer**: Secondary reviewer for the conference: The Joint Conference of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP 2021).